

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA0605

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

***CO4:** Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

QUESTIONS: 13

Total Marks: 30

Annexure A

SDG

Code of Conduct:

*I **THAMIZHARASAN S (192424087)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

THAMIZHARASAN S

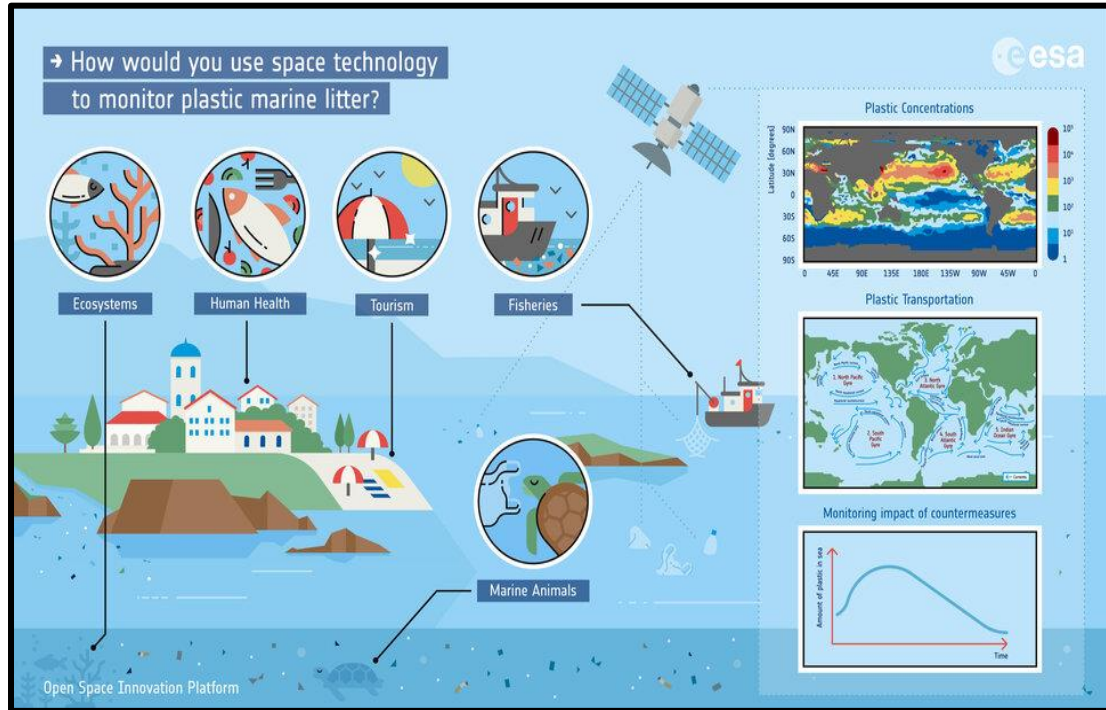
Apply KNN to classify marine pollution levels using ocean monitoring datasets

Sustainable Development Goal (SDG 14): Life Below Water

Abstract

Marine pollution is a major environmental challenge that affects ocean ecosystems, marine organisms, and water quality. Increasing levels of plastic waste, chemical contaminants, nutrient pollution, and other pollutants make continuous monitoring of marine environments essential. Artificial Intelligence (AI) and Machine Learning (ML) can help environmental agencies analyze ocean monitoring data and identify areas with different levels of pollution. This paper proposes a Machine Learning-based marine pollution monitoring system using the K-Nearest Neighbors (KNN) classification algorithm to classify marine water into "Low Pollution," "Moderate Pollution," and "High Pollution" categories. The model analyzes ocean monitoring parameters such as water temperature, pH, dissolved oxygen, salinity, turbidity, nitrate concentration, phosphate concentration, and plastic concentration to determine the pollution level of a given marine observation. Based on the classification, environmental authorities can identify polluted regions, prioritize monitoring activities, and implement appropriate pollution-control measures. The proposed system contributes to United Nations Sustainable Development Goal 14 (Life Below Water) by supporting the conservation and sustainable use of oceans and marine resources through data-driven environmental monitoring.

Keywords: Artificial Intelligence, Machine Learning, K-Nearest Neighbors, Marine Pollution, Ocean Monitoring, Water Quality, SDG 14.

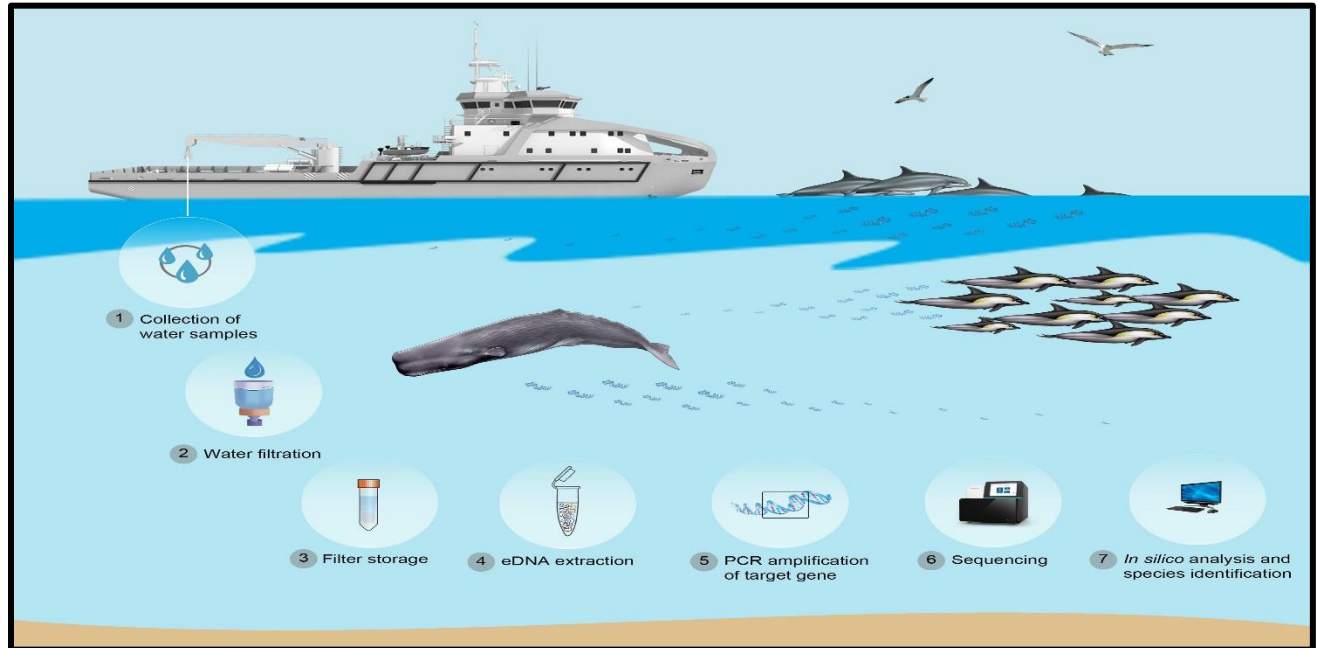


1. Problem Statement and Identification of SDG

Despite significant efforts to protect marine environments, marine pollution remains a major challenge worldwide. Oceans and coastal areas are increasingly affected by plastic waste, chemical contaminants, nutrient pollution, sewage, and other harmful substances. Traditional ocean monitoring methods often rely on manual data collection and analysis, making it difficult to identify highly polluted areas quickly and accurately.

The proposed Machine Learning-based application aims to classify marine pollution levels using K-Nearest Neighbors (KNN). By analyzing ocean monitoring data such as water temperature, pH, dissolved oxygen, salinity, turbidity, nitrate concentration, phosphate concentration, and plastic concentration, the system helps identify marine areas as "Low Pollution," "Moderate Pollution," or "High Pollution."

This work supports United Nations Sustainable Development Goal 14 (Life Below Water), which focuses on conserving and sustainably using oceans, seas, and marine resources. The proposed application contributes to protecting marine ecosystems by enabling data-driven pollution monitoring, identifying areas requiring attention, and supporting timely environmental management and pollution-control measures.



2. Background Study and Machine Learning Challenges

Ocean monitoring systems generate large amounts of environmental data, including water temperature, pH, dissolved oxygen, salinity, turbidity, nitrate concentration, phosphate concentration, and plastic pollution levels. Analyzing these data manually is time-consuming and may not effectively identify marine areas that require immediate attention. AI-based marine monitoring systems can process these datasets efficiently, identify patterns in ocean water quality, and assist environmental authorities in making informed decisions.

However, applying Machine Learning to marine pollution classification presents several challenges. Ocean monitoring datasets may contain missing or inconsistent values, reducing classification accuracy. Another challenge is the different scales of environmental parameters, such as pH, temperature, turbidity, and pollutant concentrations, which can affect distance calculations in KNN. Ocean conditions may also vary due to seasonal changes, weather, tides, and water currents, making the data complex and noisy. Additionally, selecting an appropriate value of K is important because an unsuitable K value can reduce classification performance. These challenges require proper data preprocessing, feature scaling, suitable K -value selection, and reliable ocean monitoring data for accurate marine pollution classification.

3. Learning Problem, Input Features, Target Output, and Data Representation

The proposed system uses Supervised Learning because the training data contain known pollution-level outcomes. The objective is to classify marine water into three categories: Low Pollution, Moderate Pollution, and High Pollution.

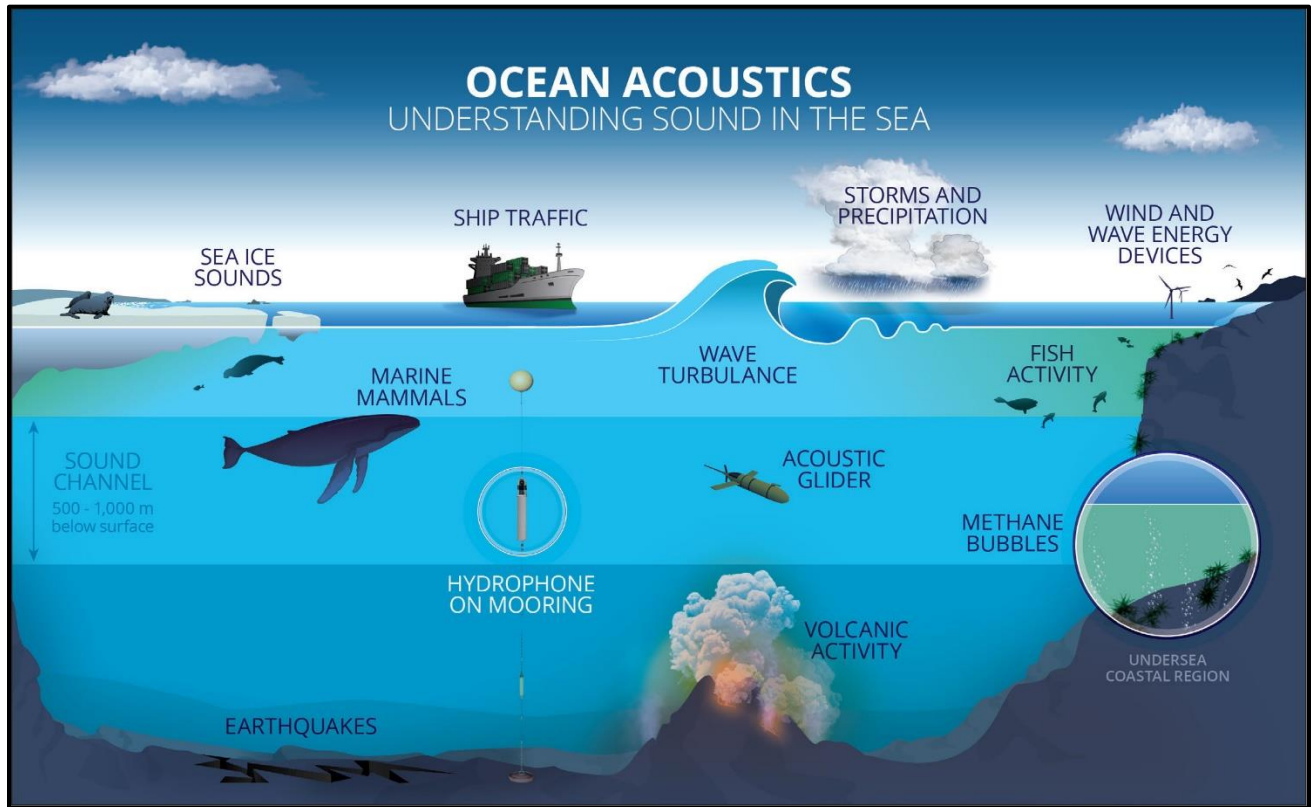
The input features include:

- Water Temperature
- pH Level
- Dissolved Oxygen
- Salinity
- Turbidity
- Nitrate Concentration
- Phosphate Concentration
- Plastic Concentration

The target output is a prediction indicating the pollution level of the marine water. The data are represented in a structured table where each row corresponds to one ocean monitoring observation and each column represents an environmental attribute.

Temperature	pH	Dissolved Oxygen	Salinity	Turbidity	Nitrate	Phosphate	Plastic Concentration	Target
27.2°C	8.1	7.2 mg/L	34.5 PSU	2.1 NTU	0.30 mg/L	0.04 mg/L	15 items/km ²	Low Pollution
28.0°C	7.9	6.5 mg/L	34.1 PSU	5.4 NTU	0.70 mg/L	0.09 mg/L	35 items/km ²	Moderate Pollution
29.1°C	7.5	4.2 mg/L	33.6 PSU	12.8 NTU	1.50 mg/L	0.18 mg/L	85 items/km ²	High Pollution

This representation enables efficient training and prediction of marine pollution levels using the K-Nearest Neighbors (KNN) Machine Learning algorithm.



4. Selection and Justification of the Machine Learning Approach

Among the Machine Learning techniques, the K-Nearest Neighbors (KNN) classification algorithm is selected because it is simple, effective, and suitable for classification problems. KNN classifies data by identifying the nearest training observations based on their feature similarity. For example, the model may compare water temperature, pH level, dissolved oxygen, turbidity, and pollutant concentrations before predicting the marine pollution level.

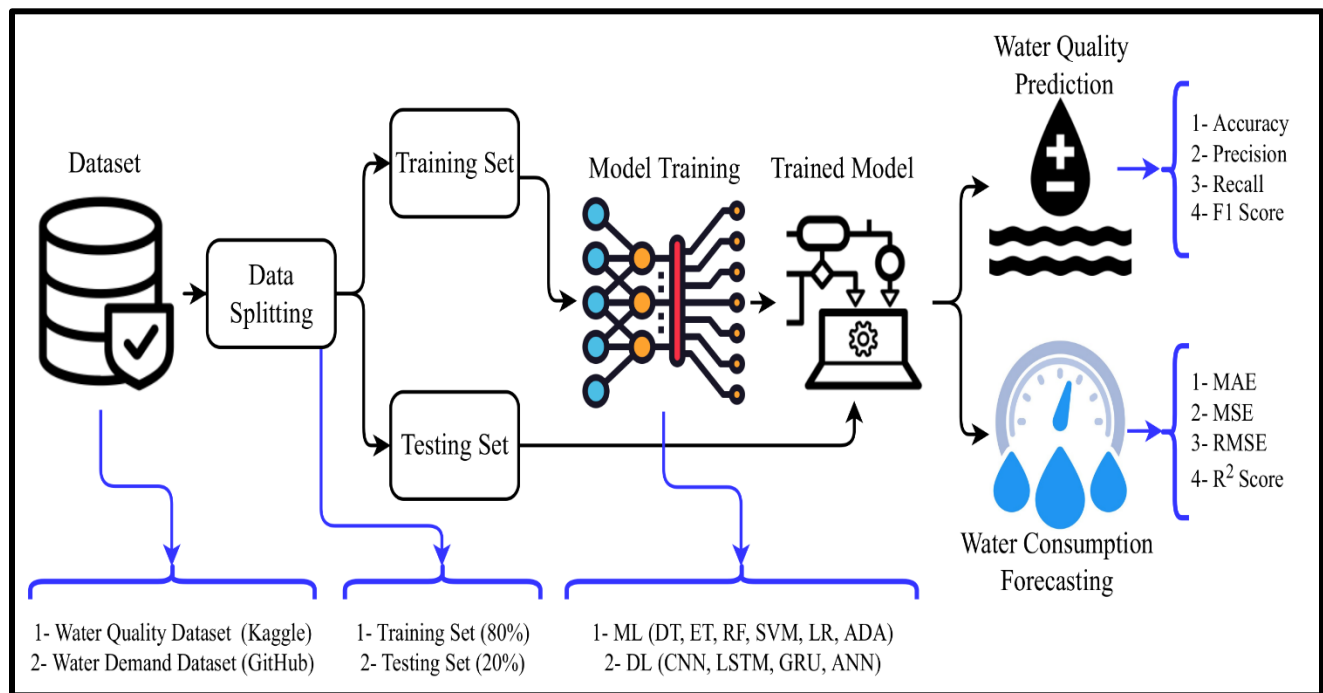
KNN is highly suitable for marine pollution classification because it can classify observations into multiple categories such as Low Pollution, Moderate Pollution, and High Pollution based on similarities in ocean monitoring data. The algorithm does not require a complex training process and can be easily implemented for environmental datasets. However, since KNN is distance-based, numerical features should be properly scaled before classification. The value of K also needs to be selected carefully because an unsuitable K value can affect prediction accuracy. Its simple classification process makes KNN useful for analyzing ocean monitoring observations and supporting data-driven marine pollution assessment.

5. Proposed Algorithm and Workflow

The proposed AI application follows a systematic workflow. First, ocean monitoring data such as water temperature, pH, dissolved oxygen, turbidity, salinity, and pollutant concentrations are collected from marine monitoring stations and environmental datasets. The data are then preprocessed to handle missing values, remove noise, and normalize numerical attributes. The cleaned dataset is divided into training and testing datasets. The K-Nearest Neighbors (KNN) algorithm analyzes the similarity between ocean monitoring samples and their neighboring data points to classify the marine pollution level. During testing, the trained KNN model predicts whether a new ocean sample has Low, Moderate, or High pollution. Finally, the system provides pollution-level information to support marine conservation, pollution control, environmental monitoring, and early intervention.

Workflow

Ocean Monitoring Data → Data Preprocessing → Feature Selection & Normalization → KNN Training → Pollution Level Classification → Environmental Monitoring & Support



This workflow enables environmental agencies and marine researchers to identify pollution levels accurately and quickly, helping them take timely actions to protect marine ecosystems.

6. Expected Outcomes, Advantages, Limitations, and Practical Applications

The proposed KNN-based marine pollution classification system is expected to identify and classify different levels of marine pollution using ocean monitoring datasets. The system can analyze parameters such as water temperature, pH, dissolved oxygen, salinity, turbidity, and pollutant concentrations. Early classification of pollution levels can help environmental authorities detect polluted areas and take appropriate actions to protect marine ecosystems.

The major advantages include simple implementation, good classification performance, easy interpretation, and flexibility. KNN does not require a complex training process and can classify new ocean monitoring observations based on their similarity to previously recorded data. This makes it suitable for analyzing environmental datasets and supporting pollution monitoring activities.

However, the system has certain limitations. KNN performance depends on the quality and size of the monitoring dataset. Noisy, incomplete, or imbalanced data may reduce classification accuracy. KNN can also require more computational time when the dataset becomes very large, since it compares new observations with existing data points.

Practical applications include marine environmental monitoring, coastal pollution assessment, ocean research organizations, fisheries management, government environmental departments, and marine conservation programs. The system can help identify pollution-prone regions, support early environmental warnings, and assist authorities in making informed decisions for protecting marine ecosystems.

7. Recommendations and Conclusion

Educational and environmental institutions should maintain accurate ocean monitoring datasets, regularly update KNN models with recent environmental data, and ensure that pollution classifications remain reliable and unbiased. Proper data preprocessing and selection of relevant features should be carried out to improve prediction accuracy. Ocean monitoring data should be securely managed, and AI-based predictions should support—not replace—expert environmental decision-making.

In conclusion, the proposed AI-based marine pollution classification system demonstrates how K-Nearest Neighbors (KNN) can be used to classify marine pollution levels using ocean monitoring datasets.

References

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